Review of Electrostatics and Steady Current

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Lecture Objective

- Provide a quick review of a topic covered in Fundamental of Electromagnetics: Electrostatics and steady current.

Maxwell's Equations

- Classical electromagnetic field theory is based on Maxwell's equations:

Table 6-1 Maxwell's equations.

Reference	Differential Form	Integral Form	
Gauss's law	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$	(6.1)
Faraday's law	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$	(6.2)*
No magnetic charges (Gauss's law for magnetism)	$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$	(6.3)
Ampère's law	$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint_C \mathbf{H} \cdot d\mathbf{l} = \int_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}\right) \cdot d\mathbf{s}$	(6.4)

*For a stationary surface S.

Question

- How is the integral form of Faraday's law written to include a moving surface?

Fundamental Vector Field Quantities in Electromagnetics

- Electric field intensity $\vec{E}$

  units = volts per meter (V/m = kg m/A/s^3)

- Electric flux density (electric displacement) $\vec{D}$

  units = coulombs per square meter (C/m^2 = A s /m^2)

- Magnetic field intensity $\vec{H}$

  units = amps per meter (A/m)

- Magnetic flux density $\vec{B}$

  units = teslas = webers per square meter (T = Wb/ m^2 = kg/A/s^3)

Relationships Between Field Intensities and Flux Densities

$$\vec{D} = \varepsilon \vec{E}$$

Permittivity of medium

$$\vec{B} = \mu \vec{H}$$

Permeability of medium

In vacuum (free space):

$$\mu_0 = 4\pi \times 10^{-7} \text{ (H/m)}$$

$$\varepsilon_0 \approx \frac{1}{36\pi} \times 10^{-9} \text{ (F/m)}$$

Simple Medium

- Constitutive parameters (permittivity, permeability, and conductivity) of a simple medium are:
  - Linear
  - Time-invariant
  - Homogeneous
  - Isotropic

Dielectric Breakdown

- Many insulating materials can be modeled as simple media.
- A dramatic violation of this model occurs when the electric field intensity exceeds the dielectric strength of the material.
- In dielectric breakdown, electrons in the valence band are detached from the molecules supporting a brief but potentially damaging impulse of current.
- Most insulating materials (other than air) are permanently damaged following dielectric breakdown.

Static Case

- Electric and magnetic fields become decoupled in the static case (i.e., there is no time variation):

$$\oint_S \vec{D} \cdot d\vec{s} = \int_V \rho_V \, dv$$

$$\oint_C \vec{E} \cdot d\vec{l} = 0$$

$$\oint_S \vec{B} \cdot d\vec{s} = 0$$

$$\oint_C \vec{H} \cdot d\vec{l} = \int_S \vec{J} \cdot d\vec{s}$$

Electrostatics

- ***Electrostatics*** is the branch of electromagnetics dealing with the effects of electric charges at rest.

$$\oint_S \vec{D} \cdot d\vec{s} = \int_V \rho_V \, dv = Q_{encl}$$

$$\oint_C \vec{E} \cdot d\vec{l} = 0$$

Electric Charge

- Static (stationary, not changing in time) ***electrical charge*** is the fundamental source of electrostatic field.
- The effects of ***electrical charge*** can be observed in the attraction/repulsion of various objects when “charged.”
- Charge comes in two varieties called “positive” and “negative.”
- Charge can occur as
  - point charges (C)
  - volume charge density (C/m^3) ⇐ **most general**
  - surface charge density (C/m^2)
  - line charge density (C/m)

Fundamental Postulates of Electrostatics

$$\nabla \cdot \vec{D} = \rho_V$$

Gauss's law

$$\nabla \times \vec{E} = 0$$

Electric field is conservative (irrotational)

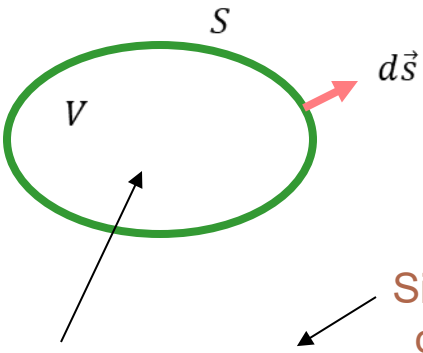
Helmholtz theorem tells us that we have now a complete specification of the electric field to within an additive constant which we assume is zero.

Gauss's Law (1 of 2)

- Gauss's law states that "the net electric flux emanating from a close surface *S* is equal to the total charge contained within the volume *V* bounded by that surface."

$$\oint_S \vec{D} \cdot d\vec{s} = Q_{encl}$$

Gauss's Law (2 of 2)



By convention, $d\vec{s}$ is taken to be outward from the volume ***V***.

$$Q_{encl} = \int_V \rho_v dv$$

Since volume charge density is the most general, we can write Q_{encl} in this way.

Electric Field of a Point Charge at the Origin in Free Space

- For a point charge at the origin in an otherwise empty universe, the electric field at any point may be derived from Gauss's law in integral form and is given by

$$\vec{E} = \hat{R}\frac{Q}{4\pi\varepsilon_0 R^2} = \frac{Q\vec{R}}{4\pi\varepsilon_0 R^3}$$

where

$$\vec{R} = \hat{x}x + \hat{y}y + \hat{z}z$$
$$R = \left|\vec{R}\right| = \sqrt{x^2 + y^2 + z^2}$$

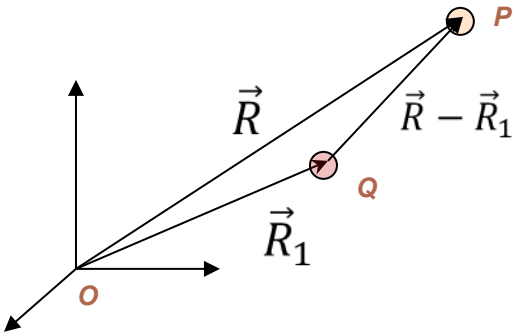
Electric Field of a Point Charge at Any Location in Free Space

- For a point charge located at a point P_1 described by a position vector given in Cartesian coordinates by

$$\vec{R}_1 = \hat{x}x_1 + \hat{y}y_1 + \hat{z}z_1$$

the electric field at P is given by

$$\vec{E} = \frac{Q\left(\vec{R} - \vec{R}_1\right)}{4\pi\varepsilon_0\left|\vec{R} - \vec{R}_1\right|^3}$$



$$\vec{R} - \vec{R}_1 = \hat{x}(x - x_1) + \hat{y}(y - y_1) + \hat{z}(z - z_1)$$
$$\left|\vec{R} - \vec{R}_1\right| = \sqrt{(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2}$$

Electric Field of a Volume Charge Density

- The electric field at P is given by

$$\vec{E} = \int_{V'} \frac{\rho_V \hat{R}'}{4\pi\varepsilon_0 R'^2} dV'$$

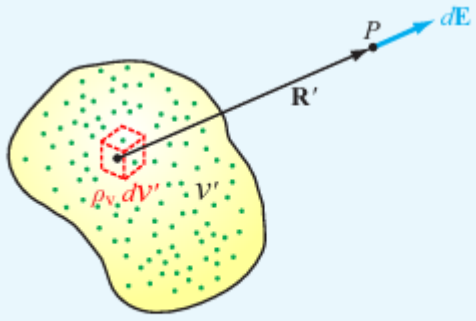


Figure 4-5 Electric field due to a volume charge distribution.

Question.

- Use the expression for the electric field due to a volume charge distribution to deduce expressions for the surface and line distributions of charge.

Electrostatic Potential (1 of 2)

- Since the electric field is *conservative*, it can be written as the gradient of a scalar field (the electrostatic potential):

$$\nabla \times \vec{E} = 0 \quad \Rightarrow \quad \vec{E} = -\nabla V$$

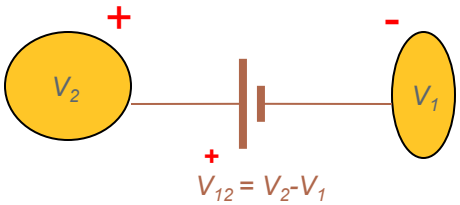
Electrostatic Potential (2 of 2)

- The ***reference point*** (P_0) is where the potential is zero (***ground*** in circuit theory).
- Often the reference is taken to be at infinity so that the potential of a point in space is defined as

$$\boxed{V(P) = -\int_{\infty}^{P} \vec{E} \cdot d\vec{l}}$$

Capacitance of a Two Conductor System (1 of 2)

- The *capacitance* of a two-conductor system is the ratio of the magnitude of the charge on one of the conductors to the magnitude of the potential difference between the two conductors.



$$C \equiv \frac{Q}{V_{12}}$$

Capacitance of a Two Conductor System (2 of 2)

- *Capacitance* is a positive quantity specified in units of *Farads*.
- *Capacitance* is the measure of the ability of a conductor configuration to store charge.

Conductors and Conductivity

- A *conductor* is a material in which electrons are free to migrate over macroscopic distances within the material.
- Metals are good conductors because they have many free electrons per unit volume.
- Other materials with a smaller number of free electrons per unit volume are also conductors.
- *Conductivity* is a measure of the ability of the material to conduct electricity.

Semiconductor

- A *semiconductor* is a material in which electrons in the outermost shell are able to migrate over macroscopic distances when a modest energy barrier is overcome.
- *Semiconductors* support the flow of both negative charges (electrons) and positive charges (holes).

Conduction Current (1 of 2)

- Conduction current results from the movement of charge carriers (electrons in a conductor, electrons and holes in a semiconductor) from atom to atom.
- The electrons that emerge from a wire are not the same electrons that entered the wire.
- Current moves through the wire with a speed much greater than the drift velocity of any individual electron.
- For simple media, the conduction current density is proportional to the electric field with the *conductivity* of the material being the constant of proportionality:

$$\boxed{\vec{J} = \sigma \vec{E}}$$

Ohm's law at a point

Conduction Current (2 of 2)

- The total current flowing through a cross-sectional area *S* is

$$\boxed{I = \int_S \bar{J} \cdot d\bar{s}}$$

- In a perfect conductor, the conductivity is infinite, but the current density must remain finite. Consequently, the electric field must be zero.
- Hence, a perfect conductor is an equipotential body; the electric potential is the same at every point in the perfect conductor.

Resistance and Resistors.

- Materials with finite conductivity provide resistance to the flow of conduction current.
- A resistor is an electrical component designed to exploit this fact.
- Ohm's law for a resistor can be derived from Ohm's law at a point:

$$V = RI$$

Current flowing through resistor

Volt drop across resistor in direction of assumed current flow

Resistance of resistor

Electromotive Force

- Steady current flow requires a closed circuit.
- Electrostatic fields produced by stationary charges are conservative. Thus, they cannot by themselves maintain a steady current flow.
- To maintain a steady current, there must be an element in the circuit wherein the potential increases in the direction of the current.
- For the potential to rise along the direction of the current, there must be a voltage source (electromotive force, *emf*) which converts some form of energy to electrical energy.

Kirchhoff's Voltage Law

- For a closed circuit containing voltages sources and resistors, we have

$$\sum V_{emf} = I \sum R$$

"the algebraic sum of the *emfs* around a closed circuit equals the algebraic sum of the voltage drops over the resistances around the circuit."

- Strictly speaking KVL only applies to circuits with steady currents (DC).
- However, for AC circuits having dimensions much smaller than a wavelength, KVL is also approximately applicable.

Summary

- When there is no time variation, Maxwell's equations separate into uncoupled pairs of equations, one describing the electric field and one pair describing the magnetic field.
- The pair of equations for electrostatics specify the curl and divergence of the electric field which are sufficient to completely specify the electric field.
- We can use Gauss's law to find the electric field for certain symmetric charge distributions.
- We can generalize the expression for the electric field due to a point charge at the origin (together with the principle of superposition) to find expressions for the electric field due to any arbitrary charge distribution.
- The capacitance of a structure is the ratio of the charge on one conductor to the potential difference between the conductors.
- Dielectric breakdown occurs when the electric field exceeds the dielectric strength of the material and determines the maximum voltage that can be placed across the terminals of a capacitor.
- The relationship between conduction current density and electric field in a material is described by Ohm's law at a point.
- KVL states that the algebraic sum of the *emfs* around a closed circuit equals the algebraic sum of the voltage drops over the resistances around the circuit.

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